

Adversarial Training for Probabilistic Robustness

Yi Zhang¹, Yuhang Chen¹, Zhen Chen², Wenjie Ruan², Xiaowei Huang²,
 Siddhartha Khastgir¹, Xingyu Zhao^{1*}

¹WMG, University of Warwick ²University of Liverpool

{yi.zhang.16,yuhang.chen.1,s.khastgir.1,xingyu.zhao}@warwick.ac.uk;

{zhen.chen2,xiaowei.huang}@liverpool.ac.uk; w.ruan@trustai.uk

Abstract

Deep learning (DL) has shown transformative potential across industries, yet its sensitivity to adversarial examples (AEs) limits its reliability and broader deployment. Research on DL robustness has developed various techniques, with adversarial training (AT) established as a leading approach to counter AEs. Traditional AT focuses on worst-case robustness (WCR), but recent work has introduced probabilistic robustness (PR), which evaluates the likelihood of AEs within a local perturbation range, providing an overall assessment of the model’s robustness and acknowledging residual risks that are more practical to manage. However, existing AT methods are fundamentally designed to improve WCR, and no dedicated methods currently target PR. To bridge this gap, we formulate a new min-max optimization as the theoretical foundation for PR-focused AT, and introduce an AT-PR training scheme with numerical algorithms to solve the new optimization problem. Our experiments, based on 70 DL models trained on common datasets and diverse architectures, demonstrate that: i) AT-PR achieves higher improvements in PR than AT-WCR methods; ii) it shows more consistent effectiveness across varying local inputs; iii) it exhibits a reduced trade-off in model’s generalization.

1. Introduction

Deep Learning (DL) has shown potential for transformative impacts across many industries including transportation, healthcare and energy. The reality that DL is not as dependable as required now becomes a major impediment to their broader deployment. Since the first discovery of the phenomenon of Adversarial Examples (AEs) for DL [21, 39, 48], *robustness* is arguably the property in the limelight, encompassing research topics like adversarial attacks in malicious contexts [7, 8, 14, 21, 36, 64, 65, 75],

AE detection during testing [19, 23, 24, 35, 42], verification with formal guarantees [20, 26, 30, 37, 44], and various robustness improvement techniques, including distillation [40], randomized smoothing [12], gradient masking [2], and the more widely used adversarial training (AT) [4, 21, 22, 36, 52]. As large language models and multi-modal foundation models gain popularity, more recent research has increasingly focused on understanding their robustness as well [11, 18, 28, 32, 33, 58, 71, 72, 74].

Regardless of specific DL tasks or model architectures, robustness broadly refers to a model’s ability to maintain consistent decisions under small input perturbations, with *local* robustness focusing on stability around individual inputs, as opposed to global robustness across the entire input distribution. More specifically, when evaluating local robustness, common problem formulations include [16, 72]: i) a binary question, “does an AE exist within a specified perturbation norm-ball?” [20, 30, 44, 63]; ii) a maximum safe distance question, “what is the largest perturbation radius within which no AE exists?” [1, 38, 60, 61]; and iii) a maximum prediction loss question, “which AE within the norm-ball yields the highest prediction loss?”, which is often applied in AT to enhance robustness [36, 56]. It is evident that the three formulations focus on *extreme scenarios*, where local robustness is evaluated based on the existence of a *deterministic* AE, reflecting a *worst-case* perspective.

A more recent and distinct view, however, considers *probabilistic robustness* (PR), employing statistical approaches to its evaluation [5, 13, 41, 49, 50, 55, 59, 60, 69, 72], it answers the question of “what is the likelihood of AEs in the given perturbation norm-ball?” [73]. However, existing AT methods are primarily designed for WCR via the well-established “min-max” optimization [3, 21, 36] (cf. Sec. 2.2), and are thus referred to as “AT-WCR” in this paper. No dedicated AT methods exist for enhancing PR, which motivates our work.

In this paper, we first formulate AT for PR as a new min-max optimization problem, and then devise AT-PR training scheme based on numerical algorithms to effectively

*Corresponding to X. Zhao, xingyu.zhao@warwick.ac.uk.

solve this training optimization. Our experiments on 70 DL models trained on common datasets and models demonstrate key strengths of AT-PR over AT-WCR: *i)* while traditional AT-WCR can also improve PR, our new AT-PR training scheme is *more effective*, especially when enhancing PR becomes challenging for traditional AT-WCR; *ii)* AT-PR not only outperforms AT-WCR in improving PR on average, but also offers *more stable and consistent* performance across different local inputs; *iii)* AT-PR has *less negative impact on model’s generalization* compared to AT-WCR.

In summary, key contributions of this paper include:

- **Problem formulation:** We lay the foundation of AT for PR by formulating a new min-max optimization problem.
- **Numerical algorithms:** We develop effective numerical algorithms to solve the AT-PR min-max optimization.
- **Empirical evaluation:** We validate the effectiveness of AT-PR through extensive experiments, showing its superior improvement in PR, more consistent effectiveness across different local inputs, and reduced impact on generalization compared to AT-WCR.
- **Open-source repository:** A repository is provided at <https://github.com/wellzline/AT-PR> with code, datasets, models, and all experiment details.

2. Preliminaries and Related Works

2.1. Worst-Case vs Probabilistic Robustness

Although definitions of robustness vary across different DL tasks and model types, it generally refers to a DL model’s ability to maintain consistent predictions despite small input perturbations. Typically it is defined as all inputs in a region η have the same prediction, where η is a small norm ball (in a L_p -norm distance) of radius γ around an input $\mathbf{x}$. A perturbed input (e.g., by adding noise on $\mathbf{x}$) $\mathbf{x}'$ within η is an AE if its prediction label differs from $\mathbf{x}$.

Evaluation methods for robustness are typically based on metrics defined around AEs [27, 75]. As discussed earlier in Sec. 1, four types of robustness metrics, as summarized in [16, 72], address: (1) a binary question, (2) a maximum safe distance question, (3) a maximum prediction loss question (Fig. 1 (a)), and (4) a statistical inference question estimating the *likelihood* of AEs within a given perturbation norm-ball (Fig. 1 (b)). The first three metrics focus on extreme scenarios, where local robustness is assessed based on the existence of a deterministic AE, reflecting a *worst-case* perspective referred to as WCR. In contrast, the fourth approach adopts a *probabilistic* view, evaluating *overall* local robustness in the presence of AEs.

PR is arguably more practical than WCR, as extreme-case focus is often unnecessary [73]. Instead, knowing the likelihood of AEs is more relevant for applications, where it is sufficient to show that the risk of AE remains below an acceptable threshold rather than exactly zero [59, 69, 72].

Inspired by [72], we illustrate the difference between the maximum loss WCR and PR in Fig. 1, omitting the binary and maximum safe distance cases as they are less relevant to the AT focus of this paper.

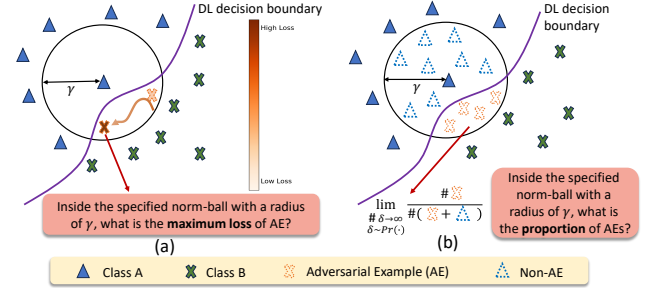


Figure 1. Worst-case (a) vs. Probabilistic (b) Robustness.

To evaluate the WCR, we normally formulate it as a question of maximizing the prediction loss, cf. Fig. 1 (a), where *adversarial attacks* are typically introduced to find such a worst-case AE with perturbation:

$$\delta = \arg \max_{\|\delta\| \leq \gamma} \mathcal{L}(\mathbf{x} + \delta; \mathbf{y}; \theta), \quad (1)$$

where δ is a small perturbation constrained within an L_p -norm ball of radius γ . This optimization causes the model to misclassify $\mathbf{x}' = \mathbf{x} + \delta$ as a different label $y' \neq y$. There are some popular attacks following this formulation, including FGSM [21], PGD [36], and ZOO [10], among others. Following common practice, we measure WCR by the success rate of some selected attacks on a test dataset. Estimation of PR is normally formalized as the statistical inference problem on the following metric:

Definition 1 (Probabilistic Robustness) For a DL classifier f_θ that takes input $\mathbf{x}$ and returns a prediction label, the PR of an input $\mathbf{x}$ in a norm ball of radius γ is:

$$PR(\mathbf{x}, \gamma) = \mathbb{E}_{\delta \sim Pr(\cdot|\mathbf{x})} [I_{\{f_\theta(\mathbf{x}+\delta)=y\}}(\mathbf{x} + \delta)], \quad (2)$$

where $I_S(\mathbf{x})$ is an indicator function—it is equal to 1 when S is true and 0 otherwise; $Pr(\cdot)$ is the local distribution of inputs representing how perturbations δ are generated, which is precisely the “input model” used by [59, 72].

Def. 1 suggests PR is the *probability* that the model prediction remains unchanged from a random perturbation $\mathbf{x}'$. A “frequentist” interpretation of this probability is: it is the *limiting relative frequency* of perturbations where the output label is preserved, in an infinite sequence of independently generated perturbations [72]. That is, the “proportion” of non-AEs in such *infinite* set of perturbations.

The work [59] is the first to define PR and introduces a black-box statistical estimator, particularly useful when

PR is extremely high (i.e., detecting AEs is a rare event). A more efficient white-box statistical estimator for PR was later proposed in [50]. Huang et al. [25] first extend PR to explainable AI (XAI). Zhang et al. [69] explore PR under functional perturbations, while the recent study [72] is the first to estimate PR in the text-to-image DL task. The recent work [73] provides an overview of PR, including its definition, estimation, enhancement and system-level integration. Without loss of generality, we adopt the estimator in [59] to estimate PR in later experiments.

2.2. Adversarial Training

AT [4, 21, 22, 36, 52] is a widely used empirical approach for improving robustness, typically formulated as a “min-max” optimization problem. It introduces a set of input perturbations δ within a norm-ball η of radius γ , designed to maximize the training loss $\mathcal{L}$. The model is then trained with these perturbations by minimizing the overall loss of the training dataset. These perturbations are often approximated using empirical adversarial attacks, such as FGSM [21] and PGD [36] due to their high efficiency. Taking a classification task as an example, the min-max optimization target of AT can be formalised as:

Definition 2 (AT for WCR) *For a DL model f_θ , trained on a dataset $\mathcal{D}$, consisting of pairs $(\mathbf{x}, y)$ where $\mathbf{x} \in \mathcal{X}$ is the input and $y \in \mathcal{Y}$ denotes the label, with loss function $\mathcal{L}(\mathbf{x}, y; \theta)$, the objective of AT-WCR can be formulated as:*

$$\min_{\theta} \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}} \left[\max_{\|\delta\| \leq \gamma} \mathcal{L}(\mathbf{x} + \delta, y; \theta) \right], \quad (3)$$

where δ represents the perturbation that maximizes classification loss $\mathcal{L}$, while the outer minimization aims to train f_θ to be robust against such adversarial perturbations.

The inner maximization is precisely the adversarial attack formulation outlined in Eq. (1). As highlighted in [3], the effectiveness of AT largely depends on identifying strong AEs within this inner maximization. Notably, the PGD attack [36] has become a critical benchmark and is widely regarded as the standard method for AT in practice [3]. Accordingly, we also use PGD in later experiments. Based on the standard min-max optimization, various AT variants have been proposed to achieve more stable and efficient training, such as [6, 15, 29, 47, 52, 57, 58, 66, 67]. Comprehensive overviews of recent AT variants can be found in surveys [3, 27], all of which fundamentally rely on the min-max problem for WCR.

In contrast, to the best of our knowledge, only three works [43, 55, 70] have explicitly aimed at improving PR [73]. Although Wang et al. [55] demonstrate that simply adding randomly perturbed inputs significantly enhances PR, the resulting AR is nearly zero. The methods of [43] and [70] propose loss functions based on common risk

measures (i.e., ρ -essential supremum, Conditional Value-at-Risk, and Entropic Value-at-Risk) to train models to minimize these risks. However, again, these approaches provide little to no improvement in AR. We argue that, akin to the well-established AT framework for AR, a dedicated AT framework for PR is necessary—one that does not compromise AR performance. Accordingly, our proposed AT-PR is the first to explicitly reformulate the min-max optimization principle for PR, while also providing comparable AR performance through practical training algorithms.

3. The Approach: AT-PR

3.1. Problem Formulation

Intuitively, the goal of AT-WCR in Def. 2 is to identify the AE that maximizes model loss and incorporate it into AT to lower the maximum loss, cf. the top-middle illustration in Fig. 2. However, to effectively improve local PR, we aim to locate the “widest peak”—the region with the highest proportion of AEs in the loss landscape (cf. the top-right illustration in Fig. 2), rather than the tallest peak. Adding the AE from this peak during AT may ideally reduce the loss for AEs around that area, thereby lowering the overall AE proportion (cf. the intuitive interpretation of PR in Def. 1). In some cases, AT-WCR and AT-PR may identify the same AE for training when the tallest peak also happens to be the widest, as shown in the second row of Fig. 2. Notably, the two cases illustrated in Fig. 2 representatively cover all possible scenarios, though additional peaks may exist in reality. Given the difference between the goals of improving WCR and PR, we formulate a new min-max problem for AT-PR:

Definition 3 (AT for PR) *For a DL model f_θ , trained on a dataset $\mathcal{D}$ consisting of pairs $(\mathbf{x}, y)$, the objective of AT-PR can be formulated as:*

$$\min_{\theta} \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}} \left[\max_{\|\delta\| \leq \gamma, PR(\mathbf{x} + \delta, k) = 0} k \right], \quad (4)$$

where k and δ are the variables in the new inner max optimization, $PR(\cdot, \cdot)$ is the metric defined in Eq. (2). As shown in Fig. 3, the intuition behind the maximization is to find an optimal δ^* that produces an AE $\mathbf{x}' = \mathbf{x} + \delta^*$ and to maximize the radius k of a smaller norm-ball centered on $\mathbf{x}'$ such that the PR within this k -sized norm-ball is 0 (i.e., all inputs within this smaller norm-ball are AEs).

The high non-convexity of DL models makes it intractable to analytically solve both the established min-max optimization in Eq. (3) and our new formulation in Eq. (4). Thus, similar to existing AT-WCR, we propose algorithms to numerically solve Eq. (4) in AT-PR in the next section.

3.2. Optimization Algorithm

While the outer minimization in Eq. (4) can be effectively handled with standard DL training methods as in AT-WCR,

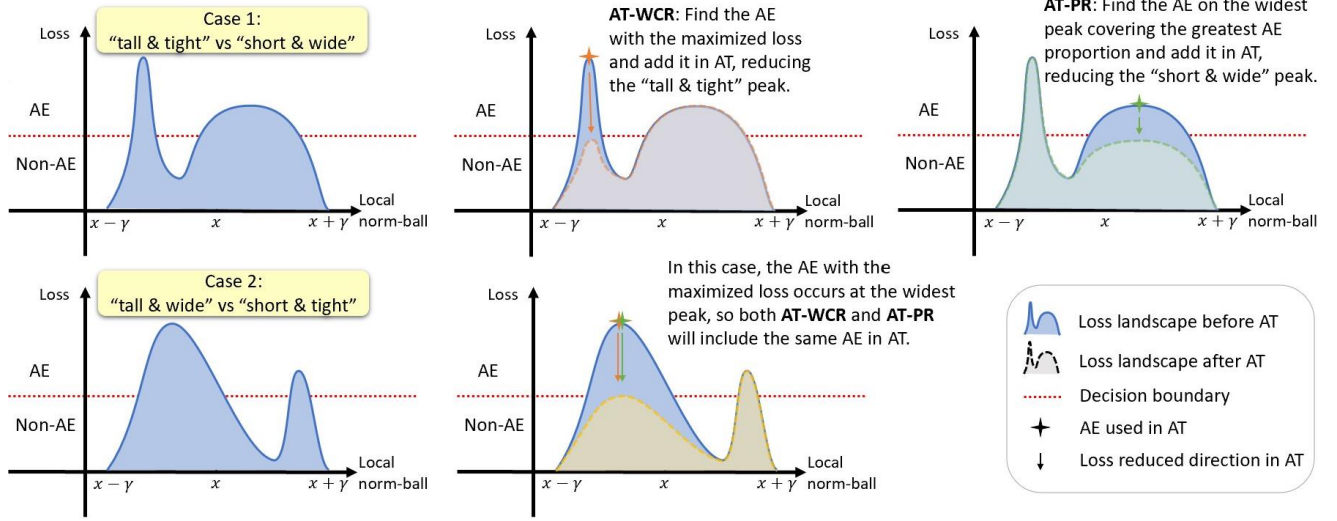


Figure 2. Two cases of the local loss landscape and their AEs identified by AT-WCR and AT-PR for AT.

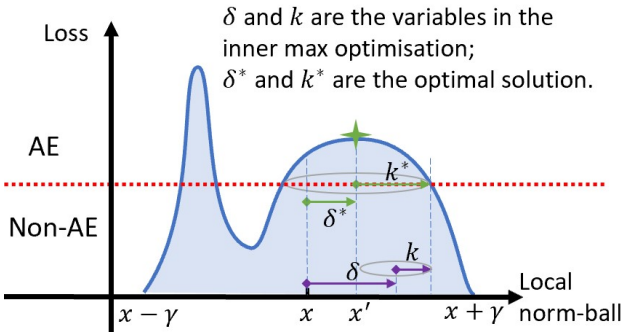


Figure 3. Illustration of the new inner-max optimization in Def. 3.

we devise the Algo. 1 to solve the inner optimization in our AT-PR. The gist of Algo. 1 is shown in Fig. 4, which essentially consists of 3 main steps:

- Step 1 (lines 1–9): Apply PGD attacks in different regions of the local norm-ball under study to find a diverse set of AEs at various local optima x'_i .
- Step 2 (lines 10–20): For each locally optimal AE found in Step 1, perform a gradient descent search to locate the nearest decision boundary.
- Step 3 (lines 21–25): Calculate the distance d_i from each x'_i to its respective decision boundary and return the AE with the largest d_i value.

Similar to AT-WCR, the returned AE will be added into the training dataset in AT-PR. We note, as an numerical algorithm to the optimization, its effectiveness may depends on certain assumptions include: *i*) the PGD attack in Step 1 may effectively find local optima by adjusting hyperparameters such as attack steps and learning rate, which aligns with the insight of “PGD will converge to local maxima”

[36]; *ii*) The gradient descent based boundary search with adaptive hyperparameters may efficiently find the decision boundary as used in some boundary attacks [9, 31, 68]. We also provide an initial theoretical analysis of the effectiveness and convergence properties of the proposed optimization algorithm in Appendix A.

Step 1: Apply PGD in different regions of the norm-ball to get a set of diverse AEs at different local optimal points x'_i .

Step 2: Perform gradient descent search from each x'_i until the nearest decision boundary.

Step 3: Calculate d_i and return the x'_i associated with the largest d_i .

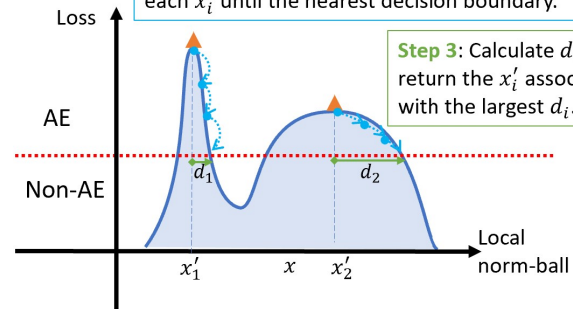


Figure 4. Illustration of the 3 main steps of Algo. 1.

We analyze the time complexity of the proposed algorithm. There are several computational steps to be considered with the assumption that the dimension of the input x is n . First, sampling α and $steps$ from uniform distributions (lines 5–6) is independent of the input and thus the time complexity is $O(1)$. Second, the PGD algorithm (line 7) consists of $steps$ gradient calculations with complexity $O(grad)^1$ and one projection calculation with complexity

¹The time complexity of gradient calculation depends on the model

Algorithm 1 PGD and gradient-based search for $\mathbf{x}'_{pr}$

Require: Inputs $[X, Y], N, \alpha_{min}, \alpha_{max}, step_{min}, step_{max}$

```
1: Initialize  $AE_s \leftarrow []$   $\triangleright$  Initialize AE candidate set
2: Apply PGD to get different AE candidates
3: for  $idx = 0$  to  $N$  do
4:    $\mathbf{x}_{init} \leftarrow \mathbf{x} + \text{Uniform}(-\gamma, \gamma)$ 
5:    $\alpha \leftarrow \text{Uniform}(\alpha_{min}, \alpha_{max})$ 
6:    $steps \leftarrow \text{Uniform}(step_{min}, step_{max})$ 
7:    $\mathbf{x}'_{idx} \leftarrow \text{pgd}(\mathbf{x}_{init}, y, \gamma, \alpha, steps)$ 
8:   Append  $\mathbf{x}'_{idx}$  to  $AE_s$ 
9: end for
Require:  $AE_s, \mathbf{x}, y, C$ 
10: Initialize  $Max\_d \leftarrow 0$ ;  $\mathbf{x}'_{pr} \leftarrow \text{None}$ 
11: for each  $\mathbf{x}' \in AE_s$  do
12:    $\tilde{\mathbf{x}} = \mathbf{x}'$ 
13:   while  $iter < C$  do
14:      $y' = f(\tilde{\mathbf{x}})$ 
15:     if  $y' = y$  then
16:       break  $\triangleright$  Exit if  $\tilde{\mathbf{x}}$  is classified correctly
17:     end if
18:      $g \leftarrow \nabla_{\tilde{\mathbf{x}}} L(\tilde{\mathbf{x}}, y)$   $\triangleright$  Compute gradient
19:      $\tilde{\mathbf{x}} = \tilde{\mathbf{x}} - \alpha \cdot g$   $\triangleright$  Update  $\tilde{\mathbf{x}}$ 
20:   end while
21:    $d \leftarrow D(\tilde{\mathbf{x}}, \mathbf{x}')$   $\triangleright$  Distance between  $\tilde{\mathbf{x}}$  &  $\mathbf{x}'$ 
22:   if  $d > Max\_d$  then
23:      $\mathbf{x}'_{pr} = \mathbf{x}'$ 
24:      $Max\_d = d$ 
25:   end if
26: end for
27: Output  $\mathbf{x}'_{pr}$   $\triangleright$  Farthest AE from decision boundary.
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$O(n)$. Third, in the effort to locate the decision boundary (lines 18–19), one gradient calculation step is performed. Last, a L_∞ distance between $\tilde{\mathbf{x}}$ and $\mathbf{x}'$ is estimated (line 21), with complexity $O(n)$. In Step 1, the algorithm uses different random starts via random initialization [46] to ensure it finds different local optima within norm-ball η (line 4), and generates one AE (lines 5–8) for each combination of input $(\mathbf{x}, y)$, radius γ , randomly sampled α and $steps$. A total of N AEs are generated. Thus the complexity upper bound of Step 1 is then $O(N * (steps * O(grad) + O(n)))$. In Step 2, with the maximum iteration number C for the nearest decision boundary locating process of N AEs, the complexity upper bound of Step 2 is then $O(C * N * O(grad))$. In Step 3, the complexity of distance estimation is $O(n)$ as discussed above. Assuming $O(grad) \gg O(n)$, which

architecture and the specific loss function employed during training. E.g., DL models with more layers or complex loss functions may require more computational resources for gradient computation due to backpropagation.

typically holds in DL scenarios, the complexity simplifies to $O(N * (steps + C) \cdot O(grad))$. As analyzed, the computational cost of AT-PR primarily comes from locating the optimal AE. However, it remains compatible with other techniques designed to enhance AT efficiency, such as [46, 62]. We provide comparative empirical results by replacing PGD with Fast-AT [62] to demonstrate such compatibility in Table 3, refer to Sec. 4 for details.

4. Experiments

We train a total of 70 DL models on four datasets: CIFAR-10, CIFAR-100, SVHN, and Tiny-ImageNet, using models such as ResNet, Wide-ResNet (WRN), Vision Transformer (ViT) [17, 34], and DeiT [51]. To evaluate AT-PR, we compare it with two standard AT-WCR (PGD and FGSM) and four advanced methods: TRADES [66], MART [57], ALP, and CLP [29]. Implementation details are in Appendix B.

4.1. Effectiveness and Efficiency of AT-PR

We assess AT-PR’s effectiveness against AT-WCR from two perspectives: 1) improvement in PR and 2) impact on generalization. We compare models trained with AT-PR, AT-WCR (FGSM and PGD) and a baseline without AT, using three metrics: (i) **Accuracy** on clean test data; (ii) **Adversarial Accuracy** under PGD, C&W, and AA attacks to reflect WCR; and (iii) **PR**, computed as $PR(\mathbf{x}, \gamma)$ for various γ using the estimator from [59]. We report the mean and standard deviation of PR on the test set, along with the generalization error (GE) to measure train–test robustness gaps.

Key findings summarized from Table 1:

i) All original models without AT show low WCR. As expected, PGD is the most effective in improving WCR, though it compromises generalization the most. AT-PR also enhances WCR, outperforming FGSM and, in some cases, even surpassing PGD, as seen with SVHN/ResNet-18, Tiny-ImageNet/ResNet-34 for PGD.Ac., and SVHN/WRN-50-2, Tiny-ImageNet/ResNet-18, ResNet-34 for CW.Ac.

ii) Both AT-WCR with FGSM and PGD can improve PR compared to the original models without AT, but they are less effective when compared to the performance of AT-PR. Specifically, the mean PR evaluations of AT-PR-trained models are higher than those trained with AT-WCR across most datasets, model architectures and norm-ball sizes. For instance, on CIFAR-100 with ResNet-18, AT-PR improves $PR(\cdot, 0.15)$ from 44.71 (FGSM) and 80.47 (PGD) to 87.38.

iii) Compared to AT-WCR, the improvement in PR becomes more pronounced as the norm-ball size γ increases, which is anticipated since larger norm-balls generally lead to less robust models. This demonstrates that our AT-PR approach is particularly useful when enhancing PR becomes more challenging for traditional AT-WCR.

iv) Each local input is associated with an estimated PR, and we report the mean and standard deviation over the test

Table 1. Evaluation results of DL models trained with different AT schemes, showing standard accuracy, adversarial accuracy (indicating WCR: PGD-10, C&W-10), and PR estimated (mean $\pm$ std) for four norm-ball sizes γ . All results are calculated over the test datasets.

Dataset	Model	AT Scheme	Accu. %	PGD Ac. %	CW Ac. %	PR($\cdot$, 0.08) %	PR($\cdot$, 0.1) %	PR($\cdot$, 0.12) %	PR($\cdot$, 0.15) %
CIFAR-10	ResNet-18	W/O AT	94.96	0.02	0.51	78.38 $\pm$ 33.14	66.3 $\pm$ 38.06	53.41 $\pm$ 40.76	37.01 $\pm$ 39.77
		AT-WCR(FGSM)	88.6	0.09	0.09	93.99 $\pm$ 17.5	88.02 $\pm$ 25.03	80.42 $\pm$ 31.81	67.77 $\pm$ 38.6
		AT-WCR(PGD)	81	45.01	44.56	97.28 $\pm$ 13.29	95.77 $\pm$ 17.07	94.1 $\pm$ 20.2	89.78 $\pm$ 26.53
		AT-PR	80.14	39.77	39.03	98.22 $\pm$ 10.38	97.28 $\pm$ 13.16	96.05 $\pm$ 16.18	93.21 $\pm$ 21.47
	WRN-50-2	W/O AT	94.14	0.23	0.14	76.96 $\pm$ 32.76	65.65 $\pm$ 37.41	54.43 $\pm$ 39.88	40.19 $\pm$ 39.49
		AT-WCR(FGSM)	92.04	0.08	0.04	95.68 $\pm$ 14.53	91.9 $\pm$ 20.25	86.67 $\pm$ 25.98	76.79 $\pm$ 33.1
		AT-WCR(PGD)	82.62	49.51	48.13	97.86 $\pm$ 12.61	97.08 $\pm$ 14.54	95.84 $\pm$ 17.14	92.62 $\pm$ 22.87
		AT-PR	82.71	43.27	43.07	98.09 $\pm$ 10.81	97.33 $\pm$ 13.03	96.22 $\pm$ 15.58	93.56 $\pm$ 20.89
CIFAR-100	ResNet-18	W/O AT	75.5	0.02	0.0	44.58 $\pm$ 42.64	32.1 $\pm$ 40.1	23.32 $\pm$ 36.41	15.62 $\pm$ 31.57
		AT-WCR(FGSM)	67.80	1.41	0.66	83.90 $\pm$ 31.11	72.89 $\pm$ 37.94	61.06 $\pm$ 42.23	44.71 $\pm$ 43.04
		AT-WCR(PGD)	52.85	19.93	19.51	95.4 $\pm$ 18.15	92.7 $\pm$ 22.72	88.98 $\pm$ 28.19	80.47 $\pm$ 36.4
		AT-PR	53.21	19.38	18.39	97.59 $\pm$ 12.96	95.96 $\pm$ 16.62	93.15 $\pm$ 21.77	87.38 $\pm$ 29.63
	WRN-50-2	W/O AT	75.46	0.49	0.45	51.11 $\pm$ 40.54	37.96 $\pm$ 39.23	28.33 $\pm$ 36.72	19.11 $\pm$ 31.97
		AT-WCR(FGSM)	73.17	0.29	0.1	85.68 $\pm$ 27.45	76.62 $\pm$ 33.89	66.37 $\pm$ 38.66	52.04 $\pm$ 41.42
		AT-WCR(PGD)	57.46	23.17	22.5	95.31 $\pm$ 17.38	92.0 $\pm$ 23.79	88.14 $\pm$ 29.15	80.67 $\pm$ 36.21
		AT-PR	59.46	22.06	21.59	96.05 $\pm$ 16.17	93.72 $\pm$ 20.47	89.48 $\pm$ 26.87	81.6 $\pm$ 34.86
SVHN	ResNet-18	W/O AT	94.96	0.08	0.12	97.11 $\pm$ 9.66	95.63 $\pm$ 11.98	93.74 $\pm$ 14.48	90.12 $\pm$ 18.40
		AT-WCR(FGSM)	93.49	4.68	5.32	98.79 $\pm$ 7.06	98.42 $\pm$ 7.70	97.86 $\pm$ 8.67	96.57 $\pm$ 10.78
		AT-WCR(PGD)	89.21	44.57	43.70	98.96 $\pm$ 5.85	98.50 $\pm$ 7.40	97.72 $\pm$ 9.61	95.41 $\pm$ 14.41
		AT-PR	90.29	45.81	42.21	99.12 $\pm$ 5.15	98.87 $\pm$ 5.93	98.54 $\pm$ 6.84	97.84 $\pm$ 8.60
	WRN-50-2	W/O AT	95.91	1.74	1.29	96.60 $\pm$ 10.36	94.77 $\pm$ 13.01	92.40 $\pm$ 15.87	87.97 $\pm$ 20.04
		AT-WCR(FGSM)	94.31	8.13	8.54	98.34 $\pm$ 7.49	97.71 $\pm$ 8.39	96.80 $\pm$ 9.69	95.12 $\pm$ 12.06
		AT-WCR(PGD)	88.91	54.99	50.58	98.72 $\pm$ 6.44	98.28 $\pm$ 7.66	97.76 $\pm$ 8.95	96.77 $\pm$ 11.18
		AT-PR	91.97	52.22	51.58	98.88 $\pm$ 6.12	98.38 $\pm$ 7.53	97.72 $\pm$ 9.19	96.26 $\pm$ 12.35
Tiny-ImageNet	ResNet-18	W/O AT	59.26	0.0	0.0	80.78 $\pm$ 33.1	72.95 $\pm$ 37.69	64.17 $\pm$ 40.91	51.14 $\pm$ 42.91
		AT-WCR(FGSM)	54.81	0.05	0.01	89.9 $\pm$ 24.97	85.33 $\pm$ 29.95	80.18 $\pm$ 33.74	69.03 $\pm$ 39.39
		AT-WCR(PGD)	47.43	21.3	19.31	96.32 $\pm$ 17.01	93.79 $\pm$ 22.1	90.16 $\pm$ 27.76	82.46 $\pm$ 35.97
		AT-PR	51.61	21.1	19.89	98.23 $\pm$ 11.06	96.1 $\pm$ 17.84	93.12 $\pm$ 23.32	86.65 $\pm$ 32.26
	ResNet-34	W/O AT	61.66	0.0	0.0	82.39 $\pm$ 31.3	75.2 $\pm$ 36.0	66.95 $\pm$ 39.42	53.7 $\pm$ 42.17
		AT-WCR(FGSM)	57	0.13	0.02	90.67 $\pm$ 23.92	85.81 $\pm$ 29.5	80.57 $\pm$ 34.13	70.22 $\pm$ 39.36
		AT-WCR(PGD)	49.33	20.97	19.58	95.45 $\pm$ 18.55	93.08 $\pm$ 23.48	89.85 $\pm$ 28.3	83.53 $\pm$ 34.99
		AT-PR	53.98	21.72	20.95	96.71 $\pm$ 15.74	95.02 $\pm$ 19.89	92.5 $\pm$ 23.87	86.23 $\pm$ 32.29

set in Table 1. Notably, AT-PR consistently achieves higher mean PR compared to models trained with AT-WCR, while also exhibiting lower standard deviations. This indicates that AT-PR not only improves PR but also offers more stable and consistent performance across different local inputs.

v) While PGD significantly improves robustness, it often reduces accuracy [66]. In contrast, AT-PR has a smaller negative impact on accuracy compared to PGD, with almost all models achieving higher accuracy than those trained using PGD while also enhancing both WCR and PR. This effect is more pronounced on larger datasets—for example, on Tiny-ImageNet with ResNet-18, AT-PR achieves an accuracy of 53.98% compared to 49.33% for PGD.

Table 2 shows that combining AT-PR with TRADES improves clean accuracy and PR on more complex attention-based models ViT (Vanilla ViT and DeiT) across different scales (DeiT-Tiny, ViT-Small, ViT-Base) for CIFAR-10. With just five epochs of AT-PR fine-tuning, all three ViT models achieve higher clean accuracy and PR while maintaining comparable WCR, showing a slight decrease on PGD-20 and AA but improved performance on CW-20.

Table 3 presents a comparison of AT-PR with four advanced AT methods (TRADES, MART, ALP and CLP) across three datasets using ResNet-18 and WRN-50-2. Re-

sults show that AT-PR consistently achieves the best PR performance across four different norm-ball sizes γ . Additionally, to verify that AT-PR is compatible with efficient AT methods [46, 62], we replace PGD with Fast-AT [62] in our framework (step 1 in Algo. 1). The result shows that *Fast-AT-PR achieves $\sim 40\%$ speedup over the original AT-PR*, with only $\sim 50\%$ extra cost compared to other AT baselines. Notably, Fast-AT-PR retains comparable PR performance to AT-PR and continues to outperform other AT baselines.

4.2. Generalization Error of AT-PR

We investigate the generalization error (GE) (train-test gap) of AT-PR to gain insights into its effectiveness. Table 4 compares the PR GE between AT-PR and PGD. Models trained with AT-PR consistently yields lower GE of up 2.2% on CIFAR-10, 2.87% on CIFAR-100, and 9.66% on Tiny-ImageNet, significantly lower than PGD’s 4.67%, 8.54%, and 11.94%, respectively. We further analyze the underlying reasons based on the *PAC-Bayes analysis for Adversarial Robustness* [54], see Appendix C for details.

4.3. AEs Used in AT-WCR vs. AT-PR

We analyze the AEs used during training to understand how AT-PR differs from AT-WCR. By comparing the AEs gen-

Table 2. ViT models trained on CIFAR-10 with TRADES and AT-PR, showing standard accuracy, adversarial accuracy (indicating WCR: PGD, C&W, AA), and PR estimated (mean $\pm$ std) for four norm-ball sizes γ .

Dataset	Model	AT Scheme	Accu. %	PGD-20. %	CW-20. %	AA. %	PR($\cdot$, 0.08) %	PR($\cdot$, 0.1) %	PR($\cdot$, 0.12) %	PR($\cdot$, 0.15) %
CIFAR-10	Deit-Ti	TRADES	79.44	49.78	46.45	45.96	96.61 $\pm$ 15.23	94.73 $\pm$ 19.36	92.69 $\pm$ 23.04	89.18 $\pm$ 27.9
		+AT-PR	81.07	49.08	46.92	45.95	97.26 $\pm$ 14.0	95.94 $\pm$ 17.12	94.16 $\pm$ 20.5	90.65 $\pm$ 25.88
	Vit-S	TRADES	81.03	50.96	48.14	47.40	96.93 $\pm$ 14.99	95.48 $\pm$ 18.4	93.58 $\pm$ 21.62	90.06 $\pm$ 26.81
		+AT-PR	83.29	50.44	48.20	47.34	97.76 $\pm$ 11.89	96.03 $\pm$ 16.46	93.98 $\pm$ 20.69	90.48 $\pm$ 26.19
	Vit-B	TRADES	83.55	53.14	50.94	49.96	97.34 $\pm$ 13.57	95.87 $\pm$ 16.99	93.78 $\pm$ 21.17	90.77 $\pm$ 25.88
		+AT-PR	86.03	52.94	51.24	49.34	97.9 $\pm$ 11.8	96.49 $\pm$ 15.2	94.44 $\pm$ 19.62	91.02 $\pm$ 25.27

Table 3. Comparison of PR($\cdot$, γ) and training time (sec/epoch) for more recent AT methods (TRADES, MART, ALP, CLP) and Fast-AT-PR.

Data	Model	Metric	AT Scheme						
			PGD	TRADES	MART	ALP	CLP	Fast-AT-PR	AT-PR
CIFAR-10	ResNet-18	PR($\cdot$, 0.08)%	97.28 $\pm$ 13.29	97.74 $\pm$ 12.42	97.4 $\pm$ 13.7	96.73 $\pm$ 15.9	97.61 $\pm$ 13.06	98.12 $\pm$ 11.52	98.22 $\pm$ 10.38
		PR($\cdot$, 0.10)%	95.77 $\pm$ 17.07	96.33 $\pm$ 16.39	95.92 $\pm$ 17.43	95.16 $\pm$ 19.53	96.29 $\pm$ 16.66	97.1 $\pm$ 14.34	97.28 $\pm$ 13.16
		PR($\cdot$, 0.12)%	94.10 $\pm$ 20.20	94.55 $\pm$ 20.31	93.93 $\pm$ 21.38	93.24 $\pm$ 23.08	94.54 $\pm$ 20.28	95.4 $\pm$ 17.90	96.05 $\pm$ 16.18
		PR($\cdot$, 0.15)%	89.78 $\pm$ 26.53	91.22 $\pm$ 25.70	90.35 $\pm$ 26.91	89.76 $\pm$ 28.14	90.86 $\pm$ 26.23	92.71 $\pm$ 23.54	93.21 $\pm$ 21.47
		Time (s/ep.)	54	61	44	64	64	97	184
	WRN-50-2	PR($\cdot$, 0.08)%	97.86 $\pm$ 12.61	97.55 $\pm$ 13.14	97.40 $\pm$ 13.61	97.75 $\pm$ 12.43	97.40 $\pm$ 13.91	97.96 $\pm$ 10.82	98.09 $\pm$ 10.81
		PR($\cdot$, 0.10)%	97.08 $\pm$ 14.54	96.14 $\pm$ 16.70	96.15 $\pm$ 16.78	96.42 $\pm$ 16.11	96.20 $\pm$ 17.58	97.22 $\pm$ 12.96	97.33 $\pm$ 13.03
		PR($\cdot$, 0.12)%	95.84 $\pm$ 17.14	94.33 $\pm$ 20.41	94.41 $\pm$ 20.37	94.64 $\pm$ 20.01	94.95 $\pm$ 19.88	96.22 $\pm$ 15.62	96.22 $\pm$ 15.58
		PR($\cdot$, 0.15)%	92.62 $\pm$ 22.87	91.01 $\pm$ 25.79	91.11 $\pm$ 25.65	91.22 $\pm$ 25.70	91.48 $\pm$ 25.69	94.74 $\pm$ 19.88	93.56 $\pm$ 20.89
		Time (s/ep.)	131	105	71	155	157	229	305
CIFAR-100	ResNet-18	PR($\cdot$, 0.08)%	95.40 $\pm$ 18.15	95.02 $\pm$ 19.27	95.45 $\pm$ 18.11	96.20 $\pm$ 16.07	96.16 $\pm$ 16.36	97.09 $\pm$ 13.48	97.59 $\pm$ 12.96
		PR($\cdot$, 0.10)%	92.70 $\pm$ 22.72	92.09 $\pm$ 24.22	92.73 $\pm$ 23.39	94.40 $\pm$ 20.08	93.81 $\pm$ 21.42	94.67 $\pm$ 18.91	95.96 $\pm$ 16.62
		PR($\cdot$, 0.12)%	88.98 $\pm$ 28.19	87.80 $\pm$ 29.80	89.35 $\pm$ 28.32	92.05 $\pm$ 24.12	91.05 $\pm$ 25.86	92.63 $\pm$ 23.91	93.15 $\pm$ 21.77
		PR($\cdot$, 0.15)%	80.47 $\pm$ 36.40	79.20 $\pm$ 37.73	81.93 $\pm$ 35.72	85.89 $\pm$ 31.65	84.97 $\pm$ 33.00	85.87 $\pm$ 31.99	87.38 $\pm$ 29.63
		Time (s/ep.)	55	61	43	64	65	97	180
	WRN-50-2	PR($\cdot$, 0.08)%	95.31 $\pm$ 17.38	95.24 $\pm$ 18.61	95.30 $\pm$ 16.87	95.34 $\pm$ 17.79	95.29 $\pm$ 17.70	96.52 $\pm$ 15.11	96.05 $\pm$ 16.17
		PR($\cdot$, 0.10)%	92.00 $\pm$ 23.79	92.30 $\pm$ 23.61	92.59 $\pm$ 22.81	92.36 $\pm$ 23.19	91.79 $\pm$ 23.87	93.49 $\pm$ 20.93	93.72 $\pm$ 20.47
		PR($\cdot$, 0.12)%	88.14 $\pm$ 29.15	88.65 $\pm$ 28.46	88.38 $\pm$ 28.65	88.51 $\pm$ 28.22	87.54 $\pm$ 29.92	88.82 $\pm$ 28.38	89.48 $\pm$ 26.87
		PR($\cdot$, 0.15)%	80.67 $\pm$ 36.21	80.35 $\pm$ 36.54	80.79 $\pm$ 36.01	80.37 $\pm$ 36.86	79.71 $\pm$ 36.76	81.02 $\pm$ 37.09	81.60 $\pm$ 34.86
		Time (s/ep.)	131	102	70	155	156	225	312
SVHN	ResNet-18	PR($\cdot$, 0.08)%	98.96 $\pm$ 5.85	99.03 $\pm$ 5.76	98.99 $\pm$ 6.15	99.05 $\pm$ 5.73	99.12 $\pm$ 5.33	98.90 $\pm$ 5.97	99.12 $\pm$ 5.15
		PR($\cdot$, 0.10)%	98.50 $\pm$ 7.40	98.46 $\pm$ 8.27	98.31 $\pm$ 8.20	98.63 $\pm$ 7.15	98.60 $\pm$ 7.21	98.53 $\pm$ 7.24	98.87 $\pm$ 5.93
		PR($\cdot$, 0.12)%	97.72 $\pm$ 9.61	97.30 $\pm$ 11.08	97.10 $\pm$ 11.26	97.85 $\pm$ 9.13	97.66 $\pm$ 9.78	98.11 $\pm$ 8.19	98.54 $\pm$ 6.84
		PR($\cdot$, 0.15)%	95.41 $\pm$ 14.41	94.15 $\pm$ 16.57	94.12 $\pm$ 16.65	95.54 $\pm$ 14.10	94.93 $\pm$ 15.00	97.10 $\pm$ 10.05	97.84 $\pm$ 8.60
		Time (s/ep.)	78	86	61	91	93	141	202
	WRN-50-2	PR($\cdot$, 0.08)%	98.72 $\pm$ 6.44	98.69 $\pm$ 6.90	98.61 $\pm$ 7.66	98.54 $\pm$ 7.52	98.59 $\pm$ 7.46	98.43 $\pm$ 6.69	98.88 $\pm$ 6.12
		PR($\cdot$, 0.10)%	98.28 $\pm$ 7.66	98.11 $\pm$ 8.66	98.07 $\pm$ 9.27	98.12 $\pm$ 8.74	98.00 $\pm$ 8.89	98.15 $\pm$ 7.29	98.38 $\pm$ 7.53
		PR($\cdot$, 0.12)%	97.76 $\pm$ 8.95	97.26 $\pm$ 10.89	97.34 $\pm$ 10.90	97.65 $\pm$ 8.85	96.98 $\pm$ 11.13	97.79 $\pm$ 8.12	97.72 $\pm$ 9.19
		PR($\cdot$, 0.15)%	96.77 $\pm$ 11.18	95.20 $\pm$ 14.84	95.45 $\pm$ 14.72	95.67 $\pm$ 14.32	94.74 $\pm$ 15.37	96.94 $\pm$ 9.80	96.26 $\pm$ 12.35
		Time (s/ep.)	190	146	102	223	225	321	420

erated by PGD and AT-PR, we find that about 10% of training inputs produce different AEs (e.g., 10.05%/8.39% for ResNet-18 and 8.83%/10.42% for WRN50-2 on CIFAR-10/CIFAR-100). This means that for roughly 90% of inputs, the loss landscapes align with Case 2 (“tall & wide vs. short & tight”) in Fig. 2, where both methods select the same AEs. However, for the remaining 10%-ish, where the landscape fits Case 1 (“tall & tight vs. short & wide”), AT-PR identifies different AEs that better optimize PR.

Fig. 5 (top row) compares the loss of AEs generated by PGD and AT-PR. For all models, PGD AEs consistently have higher losses, reflecting its objective to find AE with the maximum loss for AT. This aligns with the goal of AT-WCR, which seeks to minimize the loss on such worst-case AEs (see Fig. 2, top-middle). In contrast, AT-PR generates AEs with lower loss. Combined with the fact that our Algorithm 1 searches for examples farthest from the deci-

sion boundary, suggests that AT-PR effectively locates AEs within the widest peak of the loss landscape. Training with these AEs helps reduce the “short & wide” peak (see Fig. 2, top-right).

Fig. 5 (b) visualizes the distributions of all generated AEs $\{x'_i\}_{i=1}^n$ by analyzing the distance (L_2 norm) between each x'_i and the original input x (e.g., d_0, d_1 in Fig. 5 (a)), along with their angular relationships. The 2D coordinate in Fig. 5 (a) introduces a fixed anchor x_a near x , for an AE x'_0 , the angle θ_0 is computed between $\vec{v}_{x \rightarrow x'_0}$ and $\vec{v}_{x \rightarrow x_a}$, representing x'_0 ’s relative position, given by:

$$\theta_0 = \arccos \left(\frac{\vec{v}_{x \rightarrow x'_0} \cdot \vec{v}_{x \rightarrow x_a}}{\|\vec{v}_{x \rightarrow x'_0}\| \|\vec{v}_{x \rightarrow x_a}\|} \right).$$

The resulting angles range from approximately 88° to 92° Fig. 5 (b), reflecting the unique geometric properties of high-dimensional spaces, where random vectors highly likely to be orthogonal [53]. To further capture the variability in relative positions,

Table 4. Comparative GE experiment on DL models across different datasets. Train/test set evaluations of PR estimates (mean) for four norm-ball sizes (γ) across models trained with PGD and AT-PR, presented as $PR_{\text{train}}/PR_{\text{test}}/(PR_{\text{train}} - PR_{\text{test}})$.

Dataset	Model	AT Scheme	PR($\cdot, 0.08$) %	PR($\cdot, 0.1$) %	PR($\cdot, 0.12$) %	PR($\cdot, 0.15$) %
CIFAR-10	ResNet-18	PGD	99.02/97.28/1.74	98.36/95.77/2.59	97.33/94.1/3.23	94.45/89.78/4.67
		AT-PR	98.58/98.22/ 0.36	97.92/97.28/ 0.64	96.99/96.05/ 0.94	95.41/93.21/ 2.2
	WRN-50-2	PGD	98.88/97.86/1.01	98.19/97.08/1.1	97.49/95.84/1.64	95.3/92.62/2.67
		AT-PR	98.63/98.09/ 0.54	97.8/97.33/ 0.47	96.71/96.22/ 0.46	94.9/93.56/ 1.34
CIFAR-100	ResNet-18	PGD	99.02/95.4/3.61	97.98/92.7/5.28	95.49/88.98/6.5	89.01/80.47/8.54
		AT-PR	97.64/97.59/ 0.05	96.59/95.96/ 0.63	94.78/93.15/ 1.62	90.25/87.38/ 2.87
	WRN-50-2	PGD	98.35/95.31/3.04	96.71/92.0/4.71	93.67/88.14/5.53	86.69/80.67/6.02
		AT-PR	97.3/96.05/ 1.25	95.41/93.72/ 1.69	91.93/89.48/ 2.45	83.25/81.6/ 1.65
Tiny-ImageNet	ResNet-18	PGD	98.72/96.32/2.4	97.86/93.79/4.07	96.94/90.16/6.78	94.4/82.46/11.94
		AT-PR	99.01/98.23/ 0.78	98.02/96.1/ 1.92	96.68/93.12/ 3.56	94.31/86.65/ 7.66
	ResNet-34	PGD	98.67/95.45/3.22	98.02/93.08/6.94	96.9/89.85/7.05	93.36/83.53/9.83
		AT-PR	99.28/96.71/ 2.57	98.82/95.02/ 3.8	98.11/92.5/ 5.61	95.89/86.23/ 9.66

Table 5. PR (mean) on train/test sets, reported as: $PR_{\text{train}}/PR_{\text{test}}/(PR_{\text{train}} - PR_{\text{test}})$, for model trained with PGD, Random-AE, AT-PR.

Dataset	Model	AT Scheme	PR($\cdot, 0.08$) %	PR($\cdot, 0.1$) %	PR($\cdot, 0.12$) %	PR($\cdot, 0.15$) %
CIFAR-10	ResNet-18	PGD	99.02/97.28/1.74	98.36/95.77/2.59	97.33/94.1/3.23	94.45/89.78/4.67
		Random	98.62/97.71/0.91	97.73/96.78/0.95	96.58/95.48/1.1	94.19/92.64/1.54
		AT-PR	98.58/98.22/ 0.36	97.92/97.28/ 0.64	96.99/96.05/ 0.94	95.41/93.21/ 2.2
		PGD	99.02/95.4/3.61	97.98/92.7/5.28	95.49/88.98/6.5	89.01/80.47/8.54
CIFAR-100	ResNet-18	Random	96.6/95.87/0.73	94.85/93.77/1.08	92.18/90.92/ 1.26	85.39/85.21/ 0.18
		AT-PR	97.64/97.59/ 0.05	96.59/95.96/ 0.63	94.78/93.15/1.62	90.25/87.38/2.87

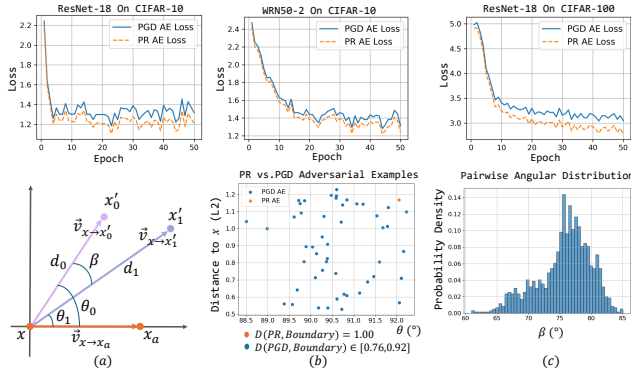


Figure 5. (Top row) Averaged loss of AEs used during AT for AT-PR and PGD. (Bottom row) (a) Illustration of angles and distances of AEs relative to x . (b) Distribution of generated AEs. (c) Pairwise angular distribution among AEs.

we computed the *Pairwise Angular Distribution* among all vectors $\{\vec{x} \rightarrow x'_i\}_{i=1}^n$. The angle β for each pair of vectors is calculated as illustrated in Fig. 5 (a), with the resulting distribution shown in Fig. 5 (c). This analysis shows that the angles between all pairs of vectors range from 60° to 85° , indicating that the generated AEs are widely spread within the norm-ball. The final AE selected by AT-PR, highlighted in orange in Fig. 5 (b), has the largest distance to nearest decision boundary (1.0 for the L2 norm), compared to other

AEs, which range from 0.76 to 0.92.

To empirically validate that AEs generated by AT-PR are most effective for improving PR, we conduct an ablation study by randomly sampling AEs from those generated in Step 1 of Algo. 1 (also illustrated as random blue spots in Fig. 5 (b)). Table 5 shows that while randomly selected AEs improve PR and GE compared to PGD, AT-PR consistently performs better. As Fig. 2 illustrates, although random AEs may fall within a short and wide peak (explaining their improved PR over PGD), they do not necessarily lie within the “widest peak” that AT-PR specifically targets. This distinction explains AT-PR’s superior PR performance.

5. Conclusion

In this work, for the first time, we introduce a novel AT method targeting PR, named as AT-PR. By reformulating a new min-max optimization, we lay the theoretical foundation for AT-PR and develop specialized numerical algorithms to solve this optimization. Extensive experimental evaluation demonstrates that AT-PR consistently outperforms traditional AT methods designed for WCR in improving PR. Additionally, we show that AT-PR incurs a smaller trade-off in model generalization compared to common AT-WCR methods.

In the future, we plan to address the challenges identified in [73], such as benchmarking PR and extending it to diverse generative AI tasks.

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